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Accounting and Financial Literacy
Level: Intermediate • Rev. 2026

Break-Even Analysis, Fixed vs Variable Costs, and Contribution Margin

Summary: Determine the exact sales volume required to cover fixed overheads. Learn how contribution margin powers profitable pricing and scenario planning.

Learning Objectives

- Differentiate between Fixed Costs and Variable Costs.
- Compute Unit Contribution Margin and Contribution Margin Ratio.
- Calculate Break-Even Point in physical units and currency value.
- Perform 'Margin of Safety' calculations to assess commercial risk.

Core Principles & Key Takeaways

- Fixed costs remain constant regardless of output; variable costs scale with unit volume.
- Contribution margin pays down fixed costs first; after break-even, it flows straight into operating profit.
- High fixed-cost structures create high operational leverage (huge profits in boom times, severe danger in downturns).

1. Cost Structures: Fixed vs Variable

All business expenditures divide into two operational categories:

* **Fixed Costs (FC):** Expenses that do not fluctuate with production volume in the short term. Examples: Office rent, core administrative salaries, property insurance, software subscription licenses, and equipment depreciation.
* **Variable Costs (VC):** Expenses that increase directly in proportion to each unit produced and delivered. Examples: Raw materials, packaging boxes, delivery courier fees, payment gateway transaction fees (e.g. 1.5% Paystack charge), and sales commission.

2. Contribution Margin Mechanics

Unit Contribution Margin (CM) is the portion of selling price remaining after covering variable costs that goes directly toward paying down fixed overheads:

$$\text{Unit CM} = \text{Selling Price per Unit (P)} - \text{Variable Cost per Unit (VC)}$$

$$\text{Contribution Margin Ratio (CMR)} = \frac{\text{Unit CM}}{\text{Selling Price (P)}} = \frac{\text{Total Contribution Margin}}{\text{Total Revenue}}$$

3. The Break-Even Formula

The **Break-Even Point (BEP)** is the operational milestone where Total Revenue equals Total Costs (Profit = 0):

$$\text{Break-Even Units (Q)} = \frac{\text{Total Fixed Costs}}{\text{Selling Price} - \text{Variable Cost per Unit}} = \frac{\text{Fixed Costs}}{\text{Unit CM}}$$

$$\text{Break-Even Revenue (₦)} = \frac{\text{Total Fixed Costs}}{\text{Contribution Margin Ratio (CMR)}}$$

4. Practical Worked Example: Commercial Bakery in Ibadan

Data:
* Selling price per loaf of specialty bread (P) = ₦1,200
* Ingredients & packaging per loaf (VC) = ₦700
* Monthly bakery rent, manager salary, and generator fuel (FC) = ₦1,500,000

Step 1: Calculate Unit Contribution Margin:
$$\text{Unit CM} = ₦1,200 - ₦700 = ₦500$$
$$\text{CMR} = \frac{₦500}{₦1,200} = 41.67\%$$

Step 2: Calculate Break-Even Units:
$$\text{Break-Even Units} = \frac{₦1,500,000}{₦500} = 3,000 \text{ loaves monthly}$$

Step 3: Calculate Break-Even Revenue:
$$\text{Break-Even Revenue} = 3,000 \times ₦1,200 = ₦3,600,000$$

Target Profit Planning:
If the bakery desires a monthly net profit of ₦1,000,000:
$$\text{Target Volume} = \frac{\text{Fixed Costs} + \text{Target Profit}}{\text{Unit CM}} = \frac{₦1,500,000 + ₦1,000,000}{₦500} = 5,000 \text{ loaves}$$

Self-Assessment & Review Questions

Q1: A digital consultancy has fixed costs of ₦2,000,000 per month. They bill clients ₦250,000 per audit and incur ₦50,000 in direct cloud and data fees per audit. How many audits must they complete to break even?

A. 8 audits
B. 10 audits
C. 15 audits
D. 20 audits

Correct Answer: B • Unit CM = ₦250,000 - ₦50,000 = ₦200,000. Break-Even Units = ₦2,000,000 / ₦200,000 = 10 audits.

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